

Protective effect of β blockers in postmenopausal women: Influence on fractures, bone density, micro and macroarchitecture

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Abstract

Introduction: Animal studies suggest that bone remodeling is under β -adrenergic control via the sympathetic nervous system. β blockers have been suggested to stimulate bone formation and/or inhibit bone resorption in animals as well as to reduce the risk of fracture in humans. The purpose of this study was to examine if these agents can have a preventive or therapeutic effect in osteoporosis.

Materials and methods: We have studied the association of β blockers use with BMD, bone geometry, microarchitecture and fractures rates in postmenopausal women referred for bone density testing. From a total sample of 944 women, we identified 158 women who were taking β blockers and 341 age-matched women as controls. Bone geometry was investigated at the femoral neck on DXA images. Microarchitecture was evaluated by the H mean fractal parameter at the calcaneus.

Results: The odds ratio for fracture (at all sites) in the β blocker users was 0.56 (95% CI, 0.30–0.99). β blocker use was associated with a higher BMD at the femoral neck (+4.2%, $p < 0.05$) and L1–L4 (+3.2%, $p < 0.05$). Proximal femur scans revealed significantly higher cortical width (+3.6%, $p < 0.05$) at the femoral neck under β blockers. Femoral shaft measurement did not significantly differ under β blockers. Medication use and lifestyle factors indicated no association between β blockers and smoking, alcohol use, physical activity, corticosteroids and estrogen therapies. The H mean parameter was significantly higher in the β blockers group (0.619 ± 0.029 vs. 0.607 ± 0.023 in controls, $p < 0.05$), suggesting a better trabecular microarchitectural organization.

Conclusion: Our data suggest that the association of current use of β blockers with low fracture risk is mediated, at least in part, by effects on BMD, cortical bone geometry and trabecular bone microarchitecture.

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Keywords: β blockers; Microarchitecture; Osteoporotic fracture; BMD

Introduction

It is estimated that up to 45% of older people suffer from cardiovascular disease and/or osteoporosis [27]. The gain in terms of public health if one therapy was able to exert beneficial effects on both heart and bone may be of great interest.

The sympathetic nervous system has been established to be involved in the mechano-transduction pathway in tail suspended rats as well as to be involved in the negative effect of

intracerebral injection of leptin on bone [21,33]. By antagonizing the sympathetic nervous system using propranolol, the bone effects (BMD and microarchitectures parameters) due to intracerebral leptin and tail suspension were decreased [33]. Furthermore, the β blocker propranolol increased bone formation in ovariectomized female rats [33] and adrenergic $\beta 2$ receptor-deficient mice demonstrate a high bone mass due to an increase in bone formation and a decrease in bone resorption parameters [10]. Therefore, β blockers have been proposed to potentially counteract loss of bone mass in postmenopausal women [8,18] but the effect on fracture risk is controversial. In two recent epidemiologic studies, β blocker use was associated with a 20–30% decrease in fracture risk [23,31], with at least

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part of the effect mediated by higher BMD. However, in the Study of Osteoporotic fractures, the association between beta-blocker use and BMD was not independent of weight and the relationship with fracture risk was inconsistent, though a persistent reduction in the risk of hip fracture was observed [28]. This inconsistency may reflect varying bone effects of the sympathetic nervous system depending on local factors such as mechanical loading, muscle mass, hormonal effects and the response of marrow cells to adipogenesis [15,36]. Furthermore, while there is incontrovertible evidence that BMD as measured by dual-energy X-ray absorptiometry (DXA) is a major determinant of the hip fracture risk [16], substantial overlap exists in BMD values between subjects with and without fractures suggesting a role for other factors such as lifestyle, propensity to falls, bone microarchitecture and femoral geometry [17,28]. Microarchitectural factors, which are considered to be relevant for explaining bone strength, are not easy to assess, the evaluation being either invasive or requiring very expensive tools. Benhamou et al. have previously validated a reproducible method to characterize the trabecular bone texture on calcaneal radiographs by fractal analysis [1,19,26]. This method has been demonstrated to discriminate patients with established osteoporosis cases from controls [25], and to a lesser extent group of corticosteroid-induced osteoporosis cases from controls [20].

The purpose of this cross-sectional study was to assess the prevalence of non-traumatic fractures in postmenopausal French women receiving β blocker treatment compared to controls and to determine the influence, if any, of potential differences in BMD, bone geometry and trabecular bone microarchitecture.

Materials and methods

Subjects

We evaluated the association between β blockers use and bone status in a sample of 944 French postmenopausal women aged between 41 and 96 years (mean 65.2 ± 9.3) recruited from routine referrals to our bone densitometry center. All women gave informed consent to participate in the study. Of the total, 158 were identified to be taking β blockers, ascertained by specific questions of exposure to β blockers (i.e. acebutolol, atenolol, celestol, celiprolol, propranolol, tomlolol, pindolol, bisoprolol, tenormine). From the remaining sample, we selected 341 age and body weight-matched women by searching for one or two age and weight-matched women for each case. The prevalence of self-reported prior fractures that had occurred after the age of 40 years was documented for cases and controls by questionnaire and checked by one of us in the medical files [3]. All spine and non-spine fractures were confirmed by radiographs. Vertebral fractures were diagnosed by vertebral deformity on radiographs according to the Genant's classification [12]. Controls and cases had a screening of vertebral fractures by Instant Vertebral Assessment (IVA) on DXA images—when the IVA image was doubtful, a plain radiograph was performed.

The following conditions were considered as exclusion criteria: osteomalacia, cancer, Paget's disease of bone, hyperparathyroidism, thyroxine treatment, renal failure, severe liver disease, long-term immobilization and incomplete medication histories. Subjects taking β blockers less than 6 months were excluded. Women receiving osteoporosis therapy (defined as more than 1 year of calcium plus vitamin D or more than 3 months of bisphosphonates, SERMs, fluoride or calcitonin), corticosteroids, HRT, statins and thiazides were included in the study as users are found frequently in the age range of our study. This

study was approved by the regional ethic's committee and entered in a Hospital Clinical Research Program.

Data

Self-reported medication use, diet and lifestyle were documented by an interview-based questionnaire. Subjects were classified as alcohol consumers if their regular intake exceeded two glasses of wine (or equivalent) per week, and as inactive if they claimed to “walk, perform normal house work or other work less than 2 hours per week”.

BMD was measured at the spine (L1–L4), the left total hip and sub-regions by dual X-ray absorptiometry (DXA, Hologic Delphi, Bedford MA). Short-term in vivo precision errors for the spine, total hip and femoral neck were 0.6%, 1.6% and 2.1%, respectively. Questionnaire completion, data collection and BMD assessments were performed concurrently.

In addition to BMD, we measured several geometric parameters from DXA data potentially involved in bone strength. We first calculated the different geometric properties of the femoral neck region (FN), a mixed cortical/trabecular site. FN diameter was obtained by dividing FN area by the constant width of the scanned region (1.5 cm). The scanned region is placed automatically by the Hologic software at the “midline” (as defined by the software) of the FN axis that runs from the lateral aspect of the greater trochanter to the inner pelvic rim (“axis” length, HAL, as defined by the software, AF; Fig. 1).

Additional manual measurements were performed on DXA scans with a program developed in our laboratory. Eight dimensions, all within the total hip, were measured as shown in Fig. 1. These comprised the intertrochanteric line (II'); the femoral neck axis length (FNAL), which is the maximum length of the FN axis, from the outer rim of the greater trochanter to the medial margin of the femoral head (AE); the greater trochanter–intertrochanter distance, from the outer rim of the greater trochanter to the intertrochanteric line (AC); the intertrochanter–pelvic rim distance, from the intertrochanteric line to the inner rim of the pelvis (CF); the intertrochanter–head distance, from the intertrochanteric line to the medial margin of the femoral head (CE); the intertrochanter–head center distance, from the intertrochanteric line to the center of the femoral head (CD) and, finally, the acetabular width (EF),

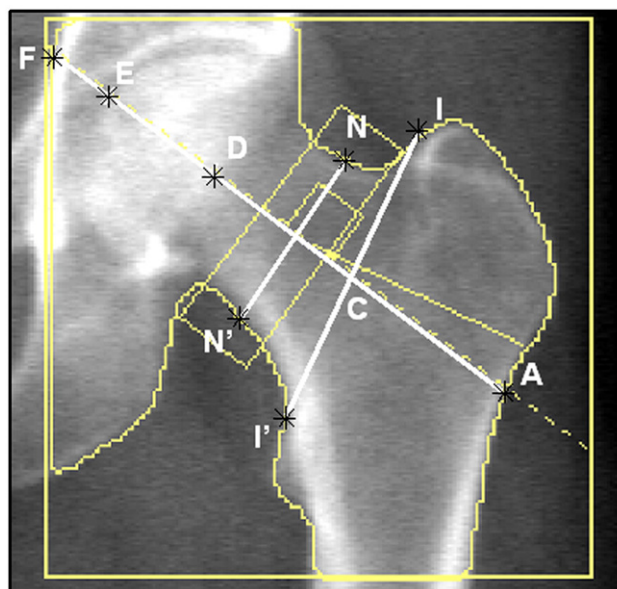


Fig. 1. Definitions of geometric parameters determined from hip DXA scans. II': intertrochanteric line; NN': neck width; AF: hip axis length is defined automatically by the DXA analysis software; AE: femoral neck axis length; AC: greater trochanter–intertrochanter distance; CF: intertrochanter–pelvic rim distance; CE: intertrochanter–inner head distance; CD: intertrochanter–head center distance; BD: shaft axis–head center distance; EF: acetabular width.

calculated as HAL minus FNAL. All measurements were performed by the main investigator, who was unaware of group assignment. The reproducibilities of the measurements were 2.8–4.7% and 4.3–7.2% for intra- and inter-observer measurements, respectively.

Microarchitecture: fractal analysis

Fractal analysis is a statistical texture analysis. It expresses the roughness of the texture and characterizes the self-similarity of the texture gray-level variations over different scales. Fractal analysis was applied on calcaneus radiographs performed with a direct digital X-ray prototype with the following parameters. The high resolution digital sensor integrated into the prototype has a 50 μm pixel size. X-ray parameters were settled at 55 kV and 20 mAs and the focal distance was established at 1.15 m. A region of interest of 256 * 256 pixels can be extracted from the image defined by anatomic marks. An oriented texture analysis was applied following 8 directions on the image. Owing its stochastic nature, the grey levels of lines in these directions were fitted to a fractal model: the Fractional Brownian Motion. For each direction the Hurst component (H) was determined by the maximum likelihood estimator. This estimation was obtained from the Fractional Gaussian Noise, an increment of the Fractional Brownian Motion. The final result for an image, H mean, is the mean value of H obtained by averaging the results of the 36 directions. The H exponent is related to the fractal dimension (D) by the relation $H = 2 - D$ [1]. The fractal analysis of calcaneal radiographs had a reproducibility of 0.099% (intraobserver CV), 0.3% (inter-observer) and 1.06% (long-term reproducibility).

Statistical analysis

All the data are shown as mean $\pm$ SEM. All the parameters were normally distributed as confirmed by the Kolmogorov–Smirnov test in the two groups. Differences between β blocker users and nonusers were evaluated using a one-way ANCOVA test. Odds ratios (OR) for fracture associated with the use of β blockers, and expressed with 95% CI, were calculated for any prior fracture, including prevalent vertebral fractures on IVA and radiographs, and wrist fractures alone. Though limited by the cross-sectional nature of the study, we also computed the odds ratio following exclusion of those fractures that had occurred prior to the self-reported date of commencing β -blocker therapy. Difference in BMD, architecture and bone geometry adjusted for potential confounders was assessed by analysis of covariance (ANCOVA). Multiple linear regressions was used to model the relationship between β -blocker and fracture with and without adjustments for potential confounders, including the use of corticosteroids, HRT, statins, thiazides and other bone active agents. Confirmatory analyses were also conducted by excluding the users of corticosteroids, HRT, statins, thiazides and other bone active agents. Finally, in order to identify the parameters that were important to distinguish β blockers users and nonusers, we performed a stepwise discriminant analysis procedure of Mahalanobis [2].

Results

Subject characteristics

Clinical data indicate that the mean duration of treatment in our population is 3348 ± 209 days with a mean dose of β blocker of 101 ± 7.3 mg.

Of the 158 β blockers users, 113 (71%) were using cardio-selective drugs. Mean age of women starting β blocker was 56 ± 11.7 years. Anthropometric measurements did not differ between β blocker users and nonusers (Table 1). As might be expected, the reported use of 3-hydroxy-3-methylglutaryl coenzyme A reductase inhibitors (“statins”) and/or thiazides was significantly higher in the β blocker group. The proportions of women using HRT, corticosteroids or osteoporosis treatment were evenly distributed between the two groups.

Table 1

Anthropometrics and lifestyle factors for women taking β blockers compared to age- and weight-matched controls (taking no β blockers) in the whole study population

	β blockers ($n=158$)	Controls ($n=341$)	p values
Age (years)	65.85 ± 9.53	64.34 ± 9.42	NS
Age at menopause (years)	50.72 ± 5.37	50.04 ± 5.07	NS
Body weight (kg)	67.10 ± 13.75	65.61 ± 12.26	NS
Body height (cm)	157.13 ± 6.51	157.84 ± 6.67	NS
Body mass index (kg/m^2)	26.73 ± 2.42	26.10 ± 2.50	NS
Cigarette smokers	5.7%	9.4%	NS
Alcohol users	13.3%	16.5%	NS
Physically active	29.7%	30.8%	NS
Reported falls during the last year	16.5%	19.9%	NS
Difficulty rising from a chair	25.3%	21.1%	NS
Statin users	25.3%	8.2%	$p < 0.001$
Thiazide users	5.1%	0.6%	$p < 0.002$
Corticosteroids users	12.3%	9.4%	NS
HRT users	45.2%	38.4%	NS
Osteoporosis therapy	17.9%	14.7%	NS

Subject are consider physically active if they claimed to walk, perform normal house work or other work less than 2 h per week.

Subjects taking HRT or osteoporosis therapy less than 3 months were considered as non treated.

Subjects taking statin and thiazide less than 6 months were excluded.

Subjects taking corticosteroids less than 1 week were excluded.

β blocker use and prevalence of fractures

A total of 156 women had experienced at least one osteoporotic fracture after the age of 40 years (including 20 hip fractures, 34 vertebral fractures, 58 wrist fractures and 44 other fractures of the foot, tibia, fibula, radius or ribs). Of these, 39 were β blocker users and 117 were in the controls giving prevalence of fracture of 24.7% vs. 34.3%, respectively ($p=0.03$) (Table 2). Using the self-reported dates of fracture and commencement of therapy, 17 of the women in the β -blocker group had sustained their fractures before exposure to the agents (fracture prevalence 13.9% in current users).

The proportion of women with fractures was lower in β blocker users across each decade of age with the most marked effects observed in women aged 70 years and older (Fig. 2) ($\chi^2=11.27$; $p=0.001$). The largest differences between the groups were observed in previous wrist and other peripheral fractures whereas the numbers of vertebral and hip fractures were small and similar in both groups (Table 2).

The prevalence of all fractures when women receiving corticosteroids, HRT, thiazides, statin or other bone therapies were excluded was 25.9% (15/58 women) and 41.3% (69/167 women) in the users and nonusers, respectively ($p=0.04$).

Association between BMD, microarchitecture and β blocker use

Compared to controls, BMD was significantly higher among β blocker users at the spine (+3%) and the femoral neck (+4%) with a statistically non-significant but similar magnitude of effect observed at the total hip (Table 2). The differences in

Table 2
Skeletal differences between users of β blocker and nonusers

	β blockers ($n=158$)	Controls ($n=341$)	p values
Subjects with fracture (s)	39 (24.7%)	117 (34.3%)	0.03
Hip fracture	6 (3.8%)	13 (3.8%)	NS
Vertebral fracture	11 (7%)	23 (6.7%)	NS
Wrist fracture	12 (7.6%)	46 (13.5%)	0.04
Other fracture	10 (6.3%)	35 (10.3%)	NS
Spine (L1–L4) BMD	0.93 ± 0.013	0.90 ± 0.008	0.01
Femoral neck BMD	0.72 ± 0.011	0.69 ± 0.006	0.02
Total hip BMD	0.85 ± 0.013	0.83 ± 0.006	0.11
H mean	0.619 ± 0.002	0.607 ± 0.001	0.003

NS: no significant; BMD: bone mineral density.

spine BMD appeared to be more marked in the more elderly women (data not shown) but no correlation was found between the duration of β blockers treatment and spine BMD L1–L4 ($r=0.10$, $p=0.86$). The H mean parameter at the calcaneus was also significantly higher in β blocker users ($+2\%$, $p<0.002$; Table 2) with no correlation between this parameter and duration of β blocker use ($r=0.005$, $p=0.97$). Adjustments for corticosteroid use, osteoporosis therapy, HRT, statin, thiazide and cigarette exposure did not alter at all the differences between the groups or the statistical significances of the differences (data not shown). Interestingly, multivariate linear regression showed no correlation between H mean and BMD at any of the measured sites (spine BMD $r=0.12$, $p=0.12$; femoral neck BMD $r=0.09$, $p=0.24$ and total hip BMD $r=0.07$, $p=0.35$) and the difference in H mean between β blocker users and controls persisted following adjustment for BMD, regardless of the site of BMD measurement ($p=0.005$).

Association between bone geometry and β blocker use

Apart from the distance between the femoral head center and the intertrochanteric line which showed a weak correlation ($r=0.12$, $p<0.05$), no significant correlations were observed between the geometric parameters and hip BMD or H mean. Small but statistically significant differences were observed between β blocker users and controls for the cortical width (Table 3). These differences were largely unaffected by

Table 3
Neck geometry from DXA scans in β blockers users and controls

	β blockers ($n=149$)	Controls ($n=322$)	p values
Femoral neck diameter (cm)	3.41 ± 0.021	3.38 ± 0.014	0.67
Cortical width of NN' (mm)	1.37 ± 0.023	1.32 ± 0.012	0.02
HAL (II')	37.90 ± 0.242	37.74 ± 0.153	0.63
FNAL (AE)	65.58 ± 0.304	65.43 ± 0.173	0.72
Greater trochanter–intertrochanter (AC)	26.48 ± 0.176	26.36 ± 0.123	0.67
Intertrochanter–pelvic rim (CF)	47.91 ± 0.336	47.68 ± 0.221	0.65
Intertrochanter–head (CE)	39.06 ± 0.287	39.08 ± 0.159	0.94
Intertrochanter–head center (CD)	22.29 ± 0.252	22.02 ± 0.190	0.61
Acetabular width (EF)	8.64 ± 0.204	8.71 ± 0.166	0.84

Bold emphasis highlights the significant differences.

For an explanation of the distance measurements, please see Fig. 1.

adjustment for concurrent statin or thiazide use or H mean but were no longer statistically significant following adjustment for differences in BMD, regardless of the skeletal site used.

To classify the skeletal parameters that were important to distinguish β blocker users and nonusers, we performed a stepwise discriminant analysis procedure using the Mahalanobis test. The correct classifications and D^2 values following addition of four significant parameters in the following order were: BMD L1–L4 (correct classification 68%, D^2 0.89), H mean (correct classification 76%, D^2 2.15) and cortical width (correct classification 82%, D^2 3.05).

OR of fracture associated with β blocker use

In univariate analysis, β blocker use was associated with a 49% reduction in fracture prevalence (any fracture), with an odds ratio of 0.51 (95% CI, 0.41–0.64, $p=0.01$) (Table 4). Given the study design, adjustments for age and weight did not alter the magnitude or significance of this association between β blocker and fracture risk for any fracture (Table 4). Furthermore, the association was unchanged following additional adjustments for statin and thiazide use or other osteoporosis medications (Table 4). Similar results were observed when the analysis was confined to wrist fractures alone (Table 4). For all fractures, the association was no longer statistically significant when adjustments were made for BMD or cortical width though the point estimates did not change substantially and the lack of significance reflected increases in the confidence limits.

Influence of HRT

An analysis of the effects of HRT was undertaken to check the validity of the study population. As expected, HRT use was associated with higher BMD at all the sites measured (e.g. mean femoral neck BMD: 0.72 g/cm^2 for HRT vs. 0.69 for nonusers of HRT). H mean was also significantly higher in HRT users. The geometric lengths were not significantly different between users of HRT and nonusers except for cortical width, which was higher in HRT group vs. nonusers. However, as underlined above, the exclusion of the HRT treated women did not suppress the significant influence of β blockers on the fractures, BMD and architectural parameters.

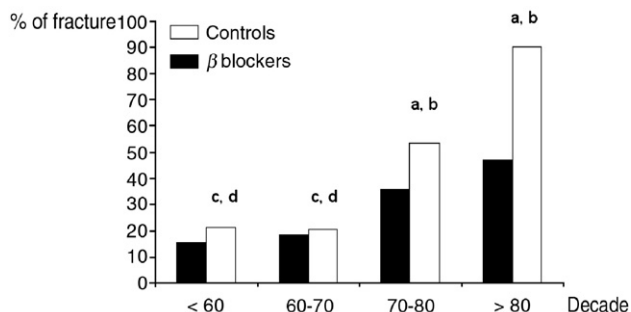


Fig. 2. Percentages of fracture in the different decade in β blockers users and nonusers. Notes a, b, c and d express significantly statistical comparisons between the different decade in the nonuser. (a) Comparison to decade <60 group ($p<0.05$). (b) Comparison to decade 60–70 ($p<0.05$). (c) Comparison to decade 70–80 ($p<0.05$). (d) Comparison to decade >80 ($p<0.05$). For the β blocker users there is no statistical difference between the decade.

Table 4
Odds ratio for prevalent fractures in β -blocker users

	Fracture type		
	Any fracture	Any fracture ^a	Wrist
Number of fracture	156 (39 are β -blocker)	139 (22 are β -blocker)	58
OR unadjusted	0.51 (0.41–0.64)	0.43 (0.26–0.70)	0.54 (0.28–1.06)
OR weight and age	0.54 (0.34–0.85)	0.35 (0.21–0.58)	0.47 (0.23–0.93)
OR adjusted for variables 1	0.54 (0.34–0.88)	0.36 (0.21–0.61)	0.40 (0.20–0.83)
OR adjusted for variables 2	0.58 (0.36–0.94)	0.39 (0.22–0.67)	0.45 (0.21–0.93)
OR adjusted for variables 3	0.32 (0.11–0.92)	0.26 (0.10–0.72)	0.19 (0.04–0.96)
OR adjusted for variables 4	0.62 (0.37–1.032)	0.59 (0.31–0.99)	0.46 (0.22–0.98)
OR adjusted for variables 5	0.63 (0.37–1.068)	0.56 (0.30–0.99)	0.56 (0.26–1.21)
OR adjusted for variables 6	0.4 (0.14–1.12)	0.29 (0.08–1.14)	0.26 (0.05–1.52)

Bold emphasis highlights the significant differences.

Variable 1: age, weight, statin, thiazide.

Variable 2: age, weight, statin, thiazide, hormonal substitution, osteoporosis therapy, corticosteroids and cigarette user.

Variable 3: variable 2 + H mean.

Variable 4: variable 2 + BMD of the entire site measured.

Variable 5: variable 2 + cortical width.

Variable 6: variable 3 + BMD of the entire site measured.

^a Restricting analysis: after exclusion of 17 women in the β -blocker group who had sustained their fractures before exposure to the agents.

Comparison of fracture vs. non fracture cases

If we compare fracture against non fracture cases, geometric results were in accordance with the literature [13]. We observed lower femoral neck axis length and lower distance between greater trochanter–intertrochanter in the fracture cases. BMD and texture data are also significantly different between these groups (data not shown). A Mahalanobis test has classified the determinants of fracture ratio: among these clinical factors, the first one is HRT use and the second one β blocker use (data not shown).

Discussion

In keeping with a growing body of literature, this study demonstrates that the use of β blocker therapy in postmenopausal women is associated with a lower prevalence of osteoporotic (OP) fractures [23,31]. For the first time, this study suggests that this lower fracture risk is independent of weight and may be mediated by beneficial effects of β blockers on BMD, cortical width and better bone microarchitecture as reflected by the H mean parameter. The validity of our results is supported by the observed effects of HRT on bone geometry which are consistent with data in the literature. Furthermore, the influence of β blocker treatment on fractures is emphasized by the fact that, among the fractures occurring under β blockers only 56% had started β blocker before the date of the fracture. Restricting the analysis to those patients with fractures reported during therapy appeared to strengthen the association between

β -blocker use and lower fracture risk. Unfortunately, the number of β blocker users was not enough to make a dose effect study as made in animal study [5]. It would be interesting to do such study in a huge database like UK General fracture Research Database.

Despite an age and weight matching between the control group and the β blocker group, we have taken age and weight into account as confounding factors. The higher concomitant use of statins and thiazides in the β blocker group could also constitute a potential bias, but our results still showed significant differences in BMD (for the lumbar spine and the femoral neck) and in H mean values at the calcaneus following adjustments for age, weight, thiazide and statin use. Moreover the lower fracture prevalence in β blocker users remained significant after these adjustments.

Furthermore, another approach (vs. adjustments) consists in excluding from the study all the subjects treated by HRT, thiazides, statin and corticosteroids as well as the anti-osteoporotic treatments. This approach considerably reduces the population (58 β blocker users vs. 167 nonusers), but the influence on fractures remains significant (OR 0.64 [95% CI, 0.31–0.84]), confirming the role of β blocker in our population.

Interestingly, the OR analyses between fracture wrist and β blocker indicate the importance of the β blocker effect on bone microarchitecture and the incidence of microarchitecture on the fracture risk. This observation is indicated by the fact that OR remains significant after adjustment with BMD measurements and by adding H mean parameter the difference between β blocker users and nonusers disappears. This observation highlights the importance of other parameters than BMD (lifestyle factors, microarchitecture properties) to characterize OP fracture risk and treatment effects. These observations support the hypothesis that the sympathetic nervous system has a catabolic effect on bone with not only a reduction of BMD but also a deterioration of trabecular microarchitecture and a change of cortical width. This has been demonstrated by a study of β agonist effects on a rat model [4]. The β blockers in animal experimentation also suggest the same mechanism: Takeda et al. have reported evidence supporting a role of propranolol (1); mice treated with this drug had increased bone mass in vertebrae secondary to increases in both bone formation rate and in osteoblast number; (2) propranolol prevented bone loss usually observed in ovariectomized mice [33].

In contrast, Dhillon et al. observed in β adrenergic receptor KO mice no protection from ovariectomy induced bone loss [9], contradicting the results of Elefteriou et al. of a protective effect of ovariectomy in $\beta 1\beta 2$ adrenergic receptor KO mice [10]. Chenu et al. suggested in sciatic denervated mice that the sympathetic nervous system did not mediate the positive effect of load-induced new cortical bone formation [7]. Togari et al. reported that adrenergic stimulation caused both anabolic and catabolic effects on bone mediated respectively by α - and β -adrenergic receptors [34].

In humans, Rejnmark et al. found no difference in BMD between β blocker users and nonusers [30]. Furthermore, they observed a higher fracture risk in subjects treated by β blockers compared to a nonuser group. However, they had a low sta-

tistical power for detecting differences in BMD due to a small number of subjects ($n=38$) and their population was 50 years old.

Recently, Reid et al. found no association between bone parameters and β blocker use after several adjustments, particularly for weight [28]. Like Reid et al., we found that once weight is taken into account some of the bone parameter differences are attenuate. But in our experience, L1–L4 and neck BMD difference remained significant after age, weight, thiazide and statin adjustments as well as H mean parameter difference. The calcaneus site has been already described to be the most sensitive site in BMD to respond to β blocker use [28]. Moreover, H mean difference remains significant after adjustments by BMD, demonstrating that H mean assesses bone microarchitecture independently of the BMD values. In addition, despite the general conclusion of Reid et al. that there is no association between fracture and β blocker use, they found an OR of 0.66 [95% CI, 0.49–0.90] for the hip fracture with adjustments by weight, age thiazide used, glucocorticoid use and alcohol use. It may be worthy of note that the proportion of patients using cardio-selective β -blockers is higher in our study than that of Reid and colleagues (71% vs. 58%). Interestingly, in a subpopulation within the latter study of patients using cardio-selective agents, Reid et al. reported a significant reduction in hip fracture and wrist fracture risk (relative risks 0.70 and 0.51, respectively) [28]. It remains to be determined whether skeletal effects may differ between classes of β -blockers.

Pasco et al. in a population of 1344 postmenopausal women, with a mean age of 70 (65 in our study), found a fracture risk reduction with β blockers [23]. The correlation between BMD and β blocker use was present in the global population and in the fracture group at the total hip and ultra distal forearm after adjusting for age and weight, but not in the group of women having not sustained a fracture. Our study brings further information since we found the same association between β blocker treatment, BMD, cortical width and H mean in the women who had not sustained a fracture. This result is not in accordance with the hypothesis that the interactions between treatment with β blocker, fracture risk and BMD exist only after the fracture episode [23].

The study of Schlienger et al. also suggests a protective effect of β blockers against OP fractures [31]. Their work was obtained on the UK General fracture Research Database, and included 30601 cases. They observed an odds ratio (OR) of fractures of 0.77 [95% CI, 0.72–0.83], while Pasco et al. observed an OR of 0.68 [95% CI, 0.49–0.96] and we had in our study an OR of 0.64 [95% CI, 0.45–0.83] by excluding the subject treated with other medication and 058 [95% CI, 0.36–0.94] after adjustments for weight, age, HRT, thiazides, statin, corticosteroids as well as anti-osteoporotic treatments [23,31].

Ferrari et al. have worked on the Framingham study and recently reported an increase in BMD under β blockers [11]. A genetic study of the β_2 adrenergic receptors polymorphism suggested that β blockers might influence bone mass in distinct genotypic subgroups [11].

More recently Turker et al. demonstrated in a prospective study higher BMD (at the lumbar vertebrae, femoral neck, trochanter, total hip) in β blocker users than nonusers of 60 to 80 years with 50 patients under β blockers treatment vs. 100 controls [35]. Furthermore, Rejnmark et al. demonstrated in a new study with a higher number of β blocker user and controls that β blocker is associated with a reduced fracture risk OR 0.91 [95% CI, 0.88–0.93] [29].

The originality of the present study is the examination of the contribution of microarchitecture effects as well as BMD to the observed differences in fracture risk. Our data suggest several mechanisms by which β blockers exert influence on bone tissue. The higher H mean parameter in the β blocker group suggests an effect of these drugs on the trabecular microarchitecture independently of the effect on BMD. This parameter obtained on radiographs has been shown to be closely related to the 3 dimensional parameters of microarchitecture [24]. This finding is in accordance with our study on a rat model: we have found that the propranolol treatment at low dose during 10 weeks induced higher trabecular number, thickness and a higher proportion of plates vs. rods [6].

Bioavailability of β blockers used clinically is in accordance with the doses previously tested in animal models [5]. In our study, several β blockers have been used by the patients; despite this heterogeneity, we have calculated the average dose divided by body weight and observed that patients received 1.53 mg/kg/day (max: 3.36 mg/kg/day, min: 0.07 mg/kg/day) of β blockers. When compared to the effective dose of propranolol in adult rats, our population is in the dose range of β blockers that has beneficial effect (0.1 mg/kg/day to 5 mg/kg/day) in this animal model [5].

The influence of bone-modifying drugs on bone geometry has been recently underlined [14,32]. Our findings on β blocker effects on the hip macro-architectural parameters show a protective effect particularly on the cortical width.

We are aware of some limitations of this study. Primarily, we have a relative small number of subjects in this study and the population is limited to women referred for bone density testing, but this population has been studied in detail, each subject having an interview with a member of our staff, and each file being reviewed particularly for fractures; furthermore, the β blocker users have been age and body weight-matched at an early step of the study. We do not have enough cases to analyze potential differences between β selective and β non-selective blockers as suggested by the β adrenergic KO mice [10]. The cross-sectional design of our study also means that it is not possible to reliably place the date of the fracture in relation to β blocker use. This, combined with the relatively small sample size, may also have contributed to the lack of any association between the BMD and H mean and duration of β blocker use. A similar lack of association has been observed in other settings. For example, in a cross-sectional study of thiazides and BMD, users were found to have higher BMD at four skeletal sites but no correlation was observed between BMD and duration of use [22]. An alternative explanation is that the protective effect has a relatively rapid onset that does not increase over time. We are not able to examine this in the current study where the average

duration of use is almost 10 years. Whether the geometric changes in bone contribute significantly to this remains a hypothesis to be tested. It is important to note that the DXA-based geometry measurements cannot be compared to real geometric measurement due to the potential magnification errors owing to the use of fan-beam DXA; the matching for weight (and similar height) in the two groups should mean that no major differences existed in the distance of the hip from the X-ray source. Finally, we can not exclude the possibility that the apparent BMD-independent improvement of trabecular architecture in β blocker users may simply be a consequence of having performed the BMD measurement at one skeletal site and the architecture measurement at another site. Nor can we determine the potential mode of action of the β blocker treatment on bone via bone formation or bone resorption. The studies on animal models are much better fitted to such an objective and suggest both a stimulation of bone formation and a reduction of bone resorption [6].

Conclusion

To our knowledge, this is the first report in humans that treatment with β blockers is associated with better micro-architectural properties as well as BMD. Our study confirms two previous studies which show less fracture cases and higher BMD with β blockers. We conclude that a prospective study of β blocker treatment should be undertaken to confirm the impact of these agents on skeletal parameters and osteoporotic fracture risk.

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